



Extra or Missing Block

Quick facilitation notes & answer key

Goal

Students debug loops with an extra inside block (does too much) or a missing inside block (job unfinished) — in block and turtle contexts — and describe how adding/removing one block fixes the result (C3.2).

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app and/or Python Turtle projected; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Run the clap, stomp, stomp loop; count 6 stomps; remove the extra block.
2. Run the forward-only square — a line; add the missing `right(90)`.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: check the inside blocks for extra/missing.

Answer key (checked by the run-gate)

Ex 1 — the second stomp is extra; Bit does 6 stomps (remove one to get 3).

Ex 2 — the turn block (`bit.right(90)`) is missing.

Ex 3 — `bit.right(90)`.

Ex 4 (challenge) — two turns of 90 = a 180 flip each side, so Bit goes back and forth; remove the extra `bit.right(90)`.

Watch for & differentiate

Common slip: changing the repeat count when a block is extra/missing.

Support: list the inside blocks and compare one-to-one with the goal.

Extend: ask what a missing forward (only turns) would draw (Bit spins in place).

Success looks like

The student spots an extra or missing inside block (not the count) and adds/removes one block to match the goal (Ex 1–3), explaining the extra-turn effect (Ex 4).